

MAT 5673 001

Partial Differential Equations I

Course Syllabus - Fall 2026



Course Information

Course Description: Prerequisite: MAT 3623 and MAT 5663, or consent of instructor. Classical theory of initial value and boundary value problems for partial differential equations, including the heat equation, wave equation, and Laplace equation, et al., and non-linear first and second order partial differential equations and calculus of variation. This course has Differential Tuition. Course Fee: GS01 \$90.

Credit Hours: 3

Course Modality: Traditional in-person

Meeting Times

Duration: 08/19/2026-12/11/2026

Campus: Main Campus

Location: [MH 3.03.12](#)

Time(s): TR 6:00 PM - 7:15 PM

Learning Goals

At the end of the course, you will be able to:

- grasp the basic theory of the classical four partial differential equations.
Outcome
grasp the basic theory of the wave equations, heat equations, Laplace equations and the Schrödinger equations.

Skills Goals

At the end of the course, you will be able to:

- grasp the theory existence, uniqueness and continuous dependence of solutions of partial differential equations; Find the solutions of Cauchy problems and initial boundary value problems.

Communicate with Me

Instructor Name: Fengxin Chen

Department
Mathematics

Office Location
Flawn 4.02.58

Student Hours
T R 5:00 pm-6:00 pm, or my appointment

Email Address: fengxin.chen@utsa.edu

Preferred Method of Communication
Use email: Fengxin.Chen@utsa.edu

About Me & My Teaching Philosophy

I am passionate about teaching and have received teaching excellence awards from two different universities. My goal as an educator is to help students experience the authentic thought processes of mathematicians as they develop, discover, and understand mathematical ideas.

Course Materials

Partial Differential Equations, s

ISBN: Softcover ISBN: 978-1-4704-6942-9 eBook ISBN: 978-1-4704-1144-2 Product Code: GSM/19.R.S.B

Authors: Lawrence C. Evan

Edition: Second Edition

To enhance your learning and adequately prepare for all assignments and assessments, please consider all course materials listed above, along with additional content provided in class, posted to our course site, or included in the Course Schedule, as required reading and study unless indicated as optional.

Assessments and Assignments

Grading Policy: Your final grade will be based on homework (20%), one midterm exam (40%), and the final exam (40%).

Homework will be assigned each class day and is expected to be completed before the next class meeting. No late or make-up work will be accepted for homework or exams, including cases of late enrollment, illness, or other personal circumstances.

To provide some flexibility, the lowest three homework/quiz scores will be dropped for every student.

If you know in advance that you will be absent on an exam day, you must make arrangements with me at least two days before the exam date so that you can take the exam before you leave.

Activities and Grading

Activity	Quantity	percent
Midterm exam	1	40
Homework	1	20
Final	1	40
Total		100%

Distribution of Course Assignments, Their Quantity, and Contribution to Final Grade.

Grade Distribution and Letter Grade

Grading Scales (by percentage):	Point Range
A+	97.5-100
A	95-97.5
A-	90-95
B+	87.5-90
B	85-87.5
B-	80-85
C+	77.5-80
C	70-75
C-	65-70
D+	65-70
D	65-67.5
D-	60-65
F	

Course Guidelines for the Use of Generative Artificial Intelligence

You are encouraged to use AI for learning purpose. Computer software and applications, including Generative Artificial Intelligence, are not allowed for exams.

Course Expectations & Policies

Reading: It is important to read the text carefully before a given section is covered in class.

Check the syllabus to determine which section will be covered. Read a section all the way through once and then come back to points of interest. I will ask you questions about the text on the quizzes.

Attendance: Attendance will not be taken but it is crucial for several reasons. First and foremost, you will gain a better understanding of the material by coming to class. Second, homework review will be given randomly during a week. In addition, changes in the syllabus or schedule may occur and will only be announced in class.

Instructor-Initiated Drops

This course uses instructor-initiated drops for students who exceed the absence and/or missed assignment limit. Therefore, up to the last day for students to withdraw from an individual course, **[date]**, you will be dropped for exceeding **[insert number of absences/assignments etc.]**. Students will receive at least one courtesy warning when approaching the absence/missed assignment limit. Notification will be sent using ASAP to the student's email address. A subsequent absence or missed assignment will result in being dropped from the course. Notification of being dropped will also be sent through ASAP to the student's email address. *This drop does not affect enrollment in other courses.* Please consult the [Dropping Courses webpage](#) for further details on the process and appeals.

Video and Audio Recording

As the instructor of this course, I may record meetings and lessons. You are expected to follow appropriate University policies and maintain the security of passwords used to access recorded lectures. Recordings may not be published, reproduced, or shared with those not in the class. If the instructor or a UTSA office plans any other uses for the recordings, consent of the students identifiable in the recordings is required before such use unless an exception is allowed by law. For more information on your privacy and class recordings, review [Student Privacy \(FERPA\) in Virtual Classrooms and Other Educational Recordings](#) and the [Guide to Secure Video Conferencing Tools](#).

Academic Freedom

Academic freedom¹ is a cornerstone of the University. Academic freedom in its teaching aspect is fundamental for the protection of the rights of the teacher in teaching and of the student to freedom in learning.² Each faculty member is entitled to full freedom in the classroom discussing the subject that the faculty member teaches.³ The University of Texas at San Antonio will not penalize or discipline members of the faculty because of their exercise of academic freedom.

Along with this freedom comes responsibility. It is the responsibility of faculty members to ensure that topics taught are related to the classroom subject. Students should be free to take reasoned exception to the data or views offered in any course of study and to reserve judgment about matters of opinion, but they are responsible for learning the content of any course of study for which they are enrolled.⁴ It is not the proper role of the university or any outside agency to attempt to shield individuals from ideas and opinions they find unwelcome, disagreeable, or even deeply offensive.⁵ Engaging with new ideas and perspectives helps students grow intellectually and is beneficial to the educational process.

- 1. Statement adapted from Texas A&M University's [Syllabus Statement Regarding Academic Freedom](#)
- 2. 1940 Statement of Principles on Academic Freedom and Tenure
- 3. Board of Regents Rule 31004; HOP Policy 4.02
- 4. American Association of University Professors Joint Statement on Rights and Freedoms of Students
- 5. The Chicago Statement

Our Commitment to Inclusivity

The University of Texas at San Antonio, a Hispanic Serving Institution situated in a global city that has been a crossroads of peoples and cultures for centuries, values diversity and inclusion in all aspects of university life. As an institution expressly founded to advance the education of Mexican Americans and other underserved communities, our university is committed to promoting access for all. UTSA, a premier public research university, fosters academic excellence through a community of dialogue, discovery and innovation that embraces the uniqueness of each voice.

Syllabus Changes

The syllabus is subject to change at the instructor’s discretion. Any changes/corrections to the course materials, assignment dates, or other updates will be communicated to the students ahead of time. You are responsible for checking Canvas for corrections or updates to the syllabus.

Course Schedule

For a list of important university-wide dates, review [One Stop’s academic calendar](#).

Due Date	Activities/Assignments	Points
9/8/26	Assignment 1	10
9/24/26	Assignment 2	10

Department Policies & Resources

Student Advocate

The student advocate is a person who investigates complaints about a specific section or instructor and attempts to resolve them through mediation. Comments or complaints sent to this person will be brought to the attention of department leadership, who will follow up. If you have questions about the logistics of your class, please contact the instructor; it is highly encouraged that you first reach out to the instructor with any issues to see if they can be resolved.

To reach the department's student advocate, please email mathematics@utsa.edu and include "Student Advocate" in the subject line of your email.

Tutoring Services

UT San Antonio provides many opportunities for student success, including tutoring services for specific math courses. The list below is subject to change based on availability.

MAT 1023, MAT 1043, MAT 1053, and MAT 1073

- We offer peer-to-peer academic support to students enrolled in a variety of courses. Both drop-in tutoring and one-on-one appointments are available.
- **Location:** [Tomás Rivera Center for Academic Excellence](#) MS 2.02.18 (behind the Rowdy statue)

MAT 1053 and MAT 1073

- Synergistic Targeted Academic Review Sessions (STARS), peer-led, interactive, and group focused sessions which cover traditionally difficult topics. The schedule will be provided by instructors through Canvas.
- **Location:** Determined by STARS tutors

MAT 1093, MAT 1213, MAT 1223, MAT 2213, MAT 2253, MAT 1313

- The Math Gym is unique in that it's a WeBWork lab designed to offer our STEM core students support from math majors and minors through targeted SI sessions. The schedule for drop-in hours and course-specific targeted sessions will be provided by instructors through Canvas.
- **Location:** [Math Gym, Main Building 0.414](#)

Note: Course-specific support for upper-division courses may be available for a particular semester. Please request this information from your instructor.

Essential Student Information

- **Important:** Bookmark and visit the [Common Syllabus Information webpage](#) to find important and valuable resources about counseling services, transitory/minor medical issues, supplemental instruction, tutoring services, academic success coaching, sexual harassment and sexual misconduct, accommodations for pregnant students, campus safety and emergency preparedness, and the Roadrunner Creed.

- For technical requirements, support, and resources, visit [Academic Innovation's Student Technical Support](#) page.
- UT San Antonio provides reasonable accommodations to students via [Student Disability Services](#). For more details on eligibility, policies, and requirements, please visit [Applying for Accommodations](#) or call (210) 458-4157.
- Your well-being is important. In addition to academic growth, life as a student can include many emotional experiences and opportunities. Being mindful of your own needs and available support can provide you with the tools to successfully navigate these experiences. Student support for well-being, including [24/7 mental health services](#), can be accessed from the [Well-being at UT San Antonio](#) site for students. If needed, we can schedule a meeting, and I can connect you to resources. You are not alone; I am available to help you find resources that might best meet your needs.
- [Student Assistance Services \(SAS\)](#) is available to help you navigate academic and personal challenges. Campus Advocates in SAS are a central point of contact for students seeking clarity on university processes and policies. They also provide a direct connection to helpful resources and advocacy rooted in care. Campus Advocates offer confidential support and personalized guidance to help students succeed.
- The [Roadrunner Pantry](#) is here to help UT San Antonio students stay healthy and focused by providing free access to groceries, toiletries, and other essentials. With locations at the Main and Downtown campuses, the pantry is organized like a small store and offers items such as canned goods, snacks, and hygiene products. Students with a valid UTSA ID can visit the pantry to pick up what they need. Our goal is to ensure every Roadrunner feels supported and has the resources to succeed. If you're ever short on food or supplies, the Roadrunner Pantry is a welcoming space you can count on.
- Students at UT San Antonio are responsible for ensuring their work is consistent with UT San Antonio's standards for academic integrity. Students should review [Section 203 of the UT San Antonio Student Code of Conduct](#) for appropriate standards of academic integrity.
- UT San Antonio provides numerous services for students from counseling to tutoring to a food pantry. Visit [Student Affairs Programs and Services](#) and [Student Success](#) for more information.
- Visit the [UT San Antonio Libraries and Museums](#) site for access to journals, research tutorials, and tech gear you can borrow and to find your department's librarian.
- Enroll in the [Roadrunner Success Playbook](#), an open-enrollment, self-paced, online hub in Canvas tailored to ensure you have the resources you need to excel at UT San Antonio.
- Follow [Digital Learning Netiquette](#) standards for your online communication activities.